

Homework 40: The Geometry of Vector Spaces

M. Usman Rashid and M. Ali

BCS All Sections

Problem 1. Affine Combination. Write $\mathbf{y}$ as affine combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

$$\mathbf{v}_1 = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -4 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 17 \\ 1 \\ 5 \end{bmatrix}.$$

- By using Theorem 1 of Section 8.1.
- By using Theorem 4 of Section 8.1.

Problem 2. Flat – A Universe Without the Zero. Explain why $x + 2y + 7z = 1$ is a Flat. Furthermore, write the dimension of the Flat.

Problem 3. Affine Independence. Show that the set S is affinely independent.

$$S = \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \end{bmatrix} \right\}.$$

- By using Theorem 5 Part c of Section 8.2.
- By using Theorem 5 Part d of Section 8.2.

Problem 4. Barycentric Coordinates. Find the Barycentric coordinates of $\mathbf{p}$ with respect to the affinely independent set S of points that precedes it.

$$S = \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \end{bmatrix} \right\}, \quad \mathbf{p} = \begin{bmatrix} 5 \\ 4 \\ -2 \\ 2 \end{bmatrix}.$$

Problem 5. Convex Combination. Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix},$$

and

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{p}_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{p}_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{p}_4 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}.$$

Let S be the orthogonal set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Determine whether each $\mathbf{p}_i$ is in $\text{Span } S$, $\text{aff } S$, and $\text{conv } S$.

Submission Date: December 6, 2024 (Friday – 11:59PM)

Good Luck